HealthCare
Smart Band

Smart Healthcare Project

Smart Band & Watch System for Fall Prevention

AI-Powered Innovation for Patient & Elderly Safety

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Global Healthcare Challenge

"Falls are a leading cause of injuries and fatalities among the elderly and post-operative patients."

WHO Statistics

28-35% of people aged 65+ experience at least one fall annually.

Problem Statement & Limitations

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- **Traditional Methods:** Colored wristbands are passive and lack real-time monitoring or instant alerts.
- **Monitoring Scale:** Large hospitals find it impossible to monitor all high-risk patients manually 24/7.
- **Severe Consequences:** Falls result in devastating physical, psychological, and economic burdens.
- **Data Gap:** Absence of automated reporting prevents medical staff from analyzing movement trends.

Solution: AI Smart Band

Introducing a groundbreaking solution for Saudi healthcare: A smart wearable powered by Artificial Intelligence (AI) and Large Language Models (LLMs).

Proactive Technology

- **Motion Sensors:** Precision balance and tilt tracking.
- **Predictive AI:** Identifying fall risk before it happens.
- **Instant Alerts:** Immediate notification to medical staff.

Key Innovation & Features

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24/7 Monitoring

Continuous real-time tracking of patient balance and physical state throughout the day.

Risk Metrics

Providing precise fall probability percentages (e.g., 75% risk) for proactive intervention.

Automated LLM Reports

Generating detailed medical summaries in natural language to explain patient status and history.

System Workflow

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Data Collection

Wearable continuously tracks movement & balance data.

AI Analysis

Instant processing to detect abnormal gait or imbalance.

Instant Alert

Immediate push notification to the nursing station.

Smart Insights

Generating periodic summaries and trend analysis.

Implementation Roadmap

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Stage	Key Activity	Expected Outcomes
Phase 1: Research	Sensor selection & initial prototype design.	Definition of balance analysis algorithms.
Phase 2: Dev	Building wearable & dashboard integration.	Seamless connectivity between device & app.
Phase 3: AI Training	Gathering trial data to train predictive models.	High accuracy & minimal false alarms.
Phase 4: Deployment	Clinical trial in a dedicated hospital ward.	Real-world performance analysis & tuning.

Technology Stack

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Intelligence

Machine Learning algorithms (XGBoost, LSTM) for high-accuracy fall prediction.

Software

Cross-platform Flutter app and a powerful React + Firebase dashboard.

Data Security

End-to-end encryption (TLS/SSL) ensuring patient privacy and compliance.

Why the Smart Band?

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35%

Annual Fall Rate (65+)

Elderly patients face increasing risks that require **proactive** rather than **reactive** solutions. Reducing this percentage by half saves millions in healthcare costs and prevents thousands of injuries.

Healthcare Impact

Reduced Hospitalization

Economic Impact

Cost

Efficiency

Patient Care

Improved Quality of Life

Expected Benefits & Outcomes

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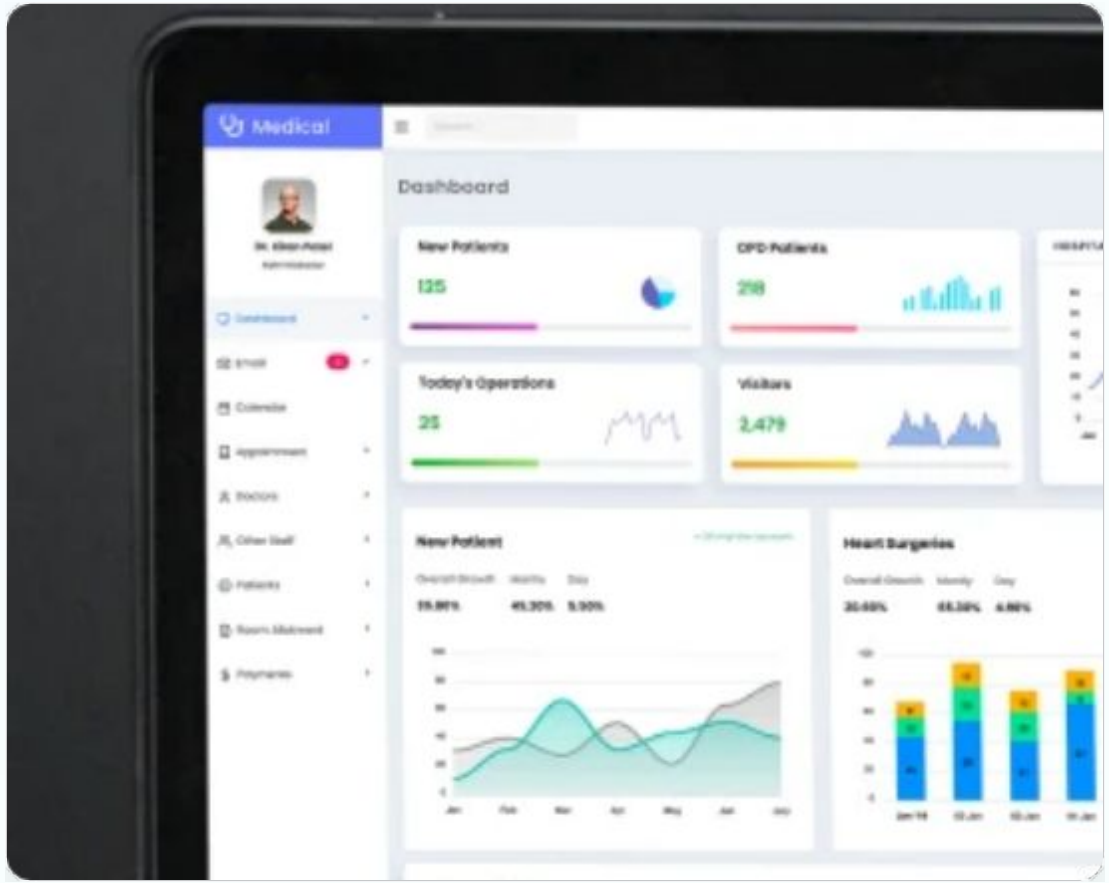

- **Enhanced Safety:** Drastic reduction in fall incidents and associated complications.
- **Staff Efficiency:** Optimizing nursing workflows through smart alerts and automated reports.
- **Cost Savings:** Minimizing the financial burden of fall-related surgery and extended care.
- **Data-Driven Decisions:** Providing clinicians with precise movement data for better recovery plans.

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Future Vision

"Integrating our system with hospital vitals—monitoring glucose, blood pressure, and heart rate—for a holistic, predictive patient profile."

Leading the Digital Transformation in Saudi Healthcare

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Questions?

Thank you for your attention

Smart Band Project for Fall Prevention

Innovating for a Safer Future